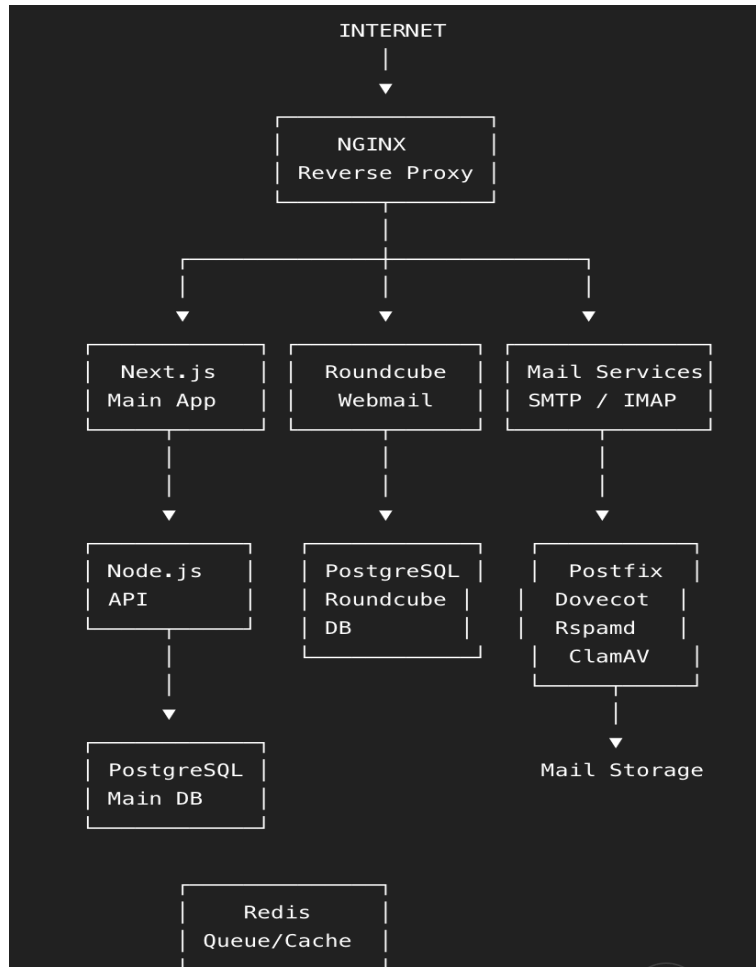


Nextjs + Roundcube Webmail



Important: Roundcube ko Next.js ke andar convert/merge ni krna hai.

Next.js handle karega	Roundcube handle karega
• Login/Register • Tenant/company • Admin dashboard • Domain management • User management • Mailbox creation • Plans • Billing • Subscription • DNS verification • DKIM information • SPF information • DMARC information • Calendar UI	• Inbox • Sent • Drafts • Trash • Spam • Compose • Reply • Reply all • Forward • Attachments • MIME rendering • IMAP folders • Message flags • Search

• Contacts UI • Security settings • 2FA • Audit logs • Notifications • API • User settings	• Address book/basic mail contacts • Mail reading • Mail sending
--	--

Roundcube is specifically an IMAP webmail client and has a plugin/skin system, so custom mail-side features ko plugin layer mein rakhna better rahega

- Next.js ke current setup ke liye Node.js 20.9+ required hai according to current official docs.
- https://github.com/roundcube/roundcubemail/blob/master/config/config.inc.php.sample?utm_source=chatgpt.com

Mail server stack

Ye bahut important hai.

SMTP

Postfix

IMAP

Dovecot

Spam

Rspamd

Virus

ClamAV

DKIM

OpenDKIM

ya Rspamd DKIM signing.

Rspamd open-source Apache 2.0 licensed hai aur commercial use permitted hai

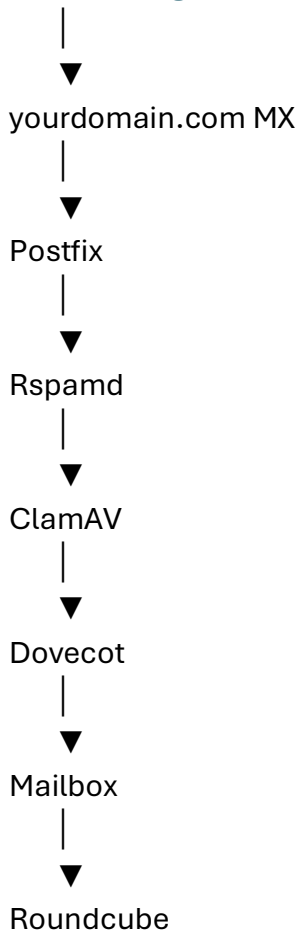
https://docs.rspamd.com/?utm_source=chatgpt.com

Mail flow

Incoming email

Suppose:

customer@gmail.com

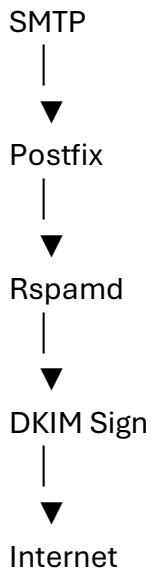


User Roundcube mein inbox dekhaga.

Outgoing email

Roundcube





Next.js → Roundcube integration

User:

<https://mail.yourcompany.com>

Next.js:

<https://app.yourcompany.com>

Admin:

<https://admin.yourcompany.com>

API:

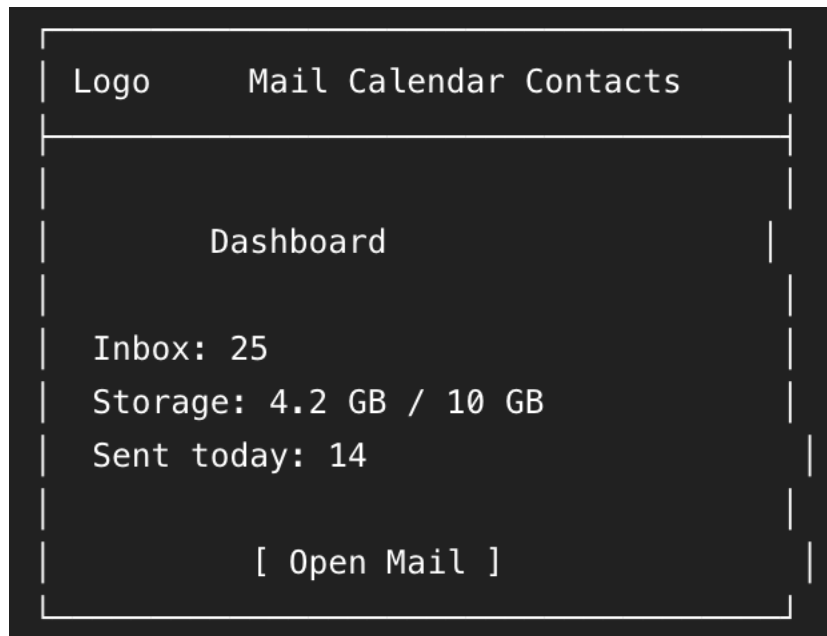
<https://api.yourcompany.com>

Roundcube:

<https://webmail.yourcompany.com>

Lekin user ko ideally ye difference dikhe hi nahi.

Next.js dashboard:



Click:

Open Mail



webmail.yourdomain.com



Roundcube

Login kaise manage karna hai?

Ye carefully design karna hai.

Next.js:

Login



Node API



PostgreSQL



JWT/Session

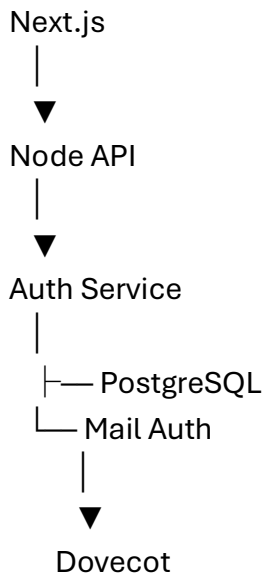
Mail credentials:

user@company.com

password

Dovecot/Postfix authentication ke liye **same user credentials** use honge.

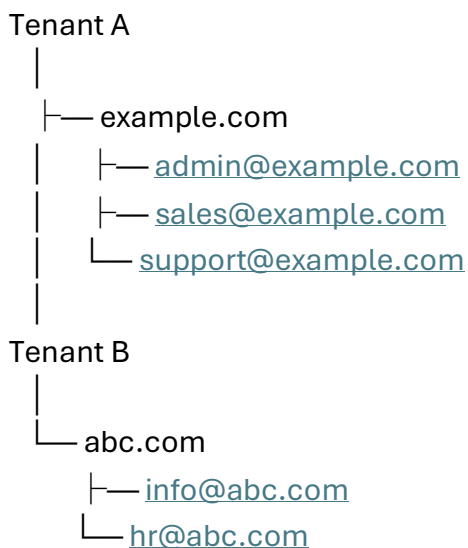
Better future architecture:



Company / Tenant structure

Agar SaaS banana hai to beginning se multi-tenancy rakho

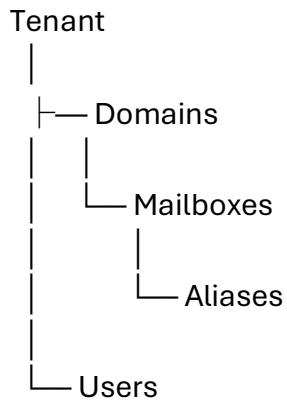
Example:



DB:

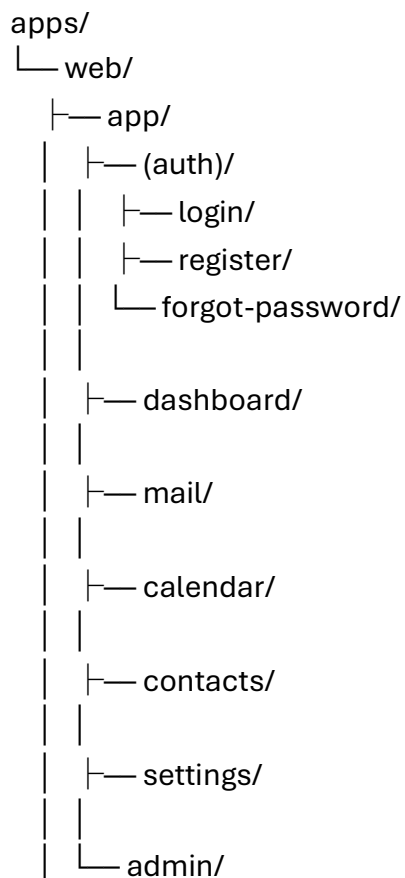
tenants
domains
users
mailboxes
aliases

Relationship:



Project structure

Next.js



```
|
|
|└─ components/
|└─ features/
|  |└─ auth/
|  |└─ domains/
|  |└─ mailboxes/
|  |└─ billing/
|  |└─ calendar/
|  └─ contacts/
|
|└─ lib/
|└─ hooks/
|└─ services/
|└─ types/
└─ middleware.ts
```

Node/NestJS structure

```
apps/
└─ api/
  |└─ src/
  |  |└─ modules/
  |  |
  |  |└─ auth/
  |  |└─ users/
  |  |└─ tenants/
  |  |└─ domains/
  |  |└─ mailboxes/
  |  |└─ aliases/
  |  |└─ dns/
  |  |└─ dkim/
  |  |└─ security/
  |  |└─ two-factor/
  |  |└─ migration/
  |  |└─ calendar/
  |  |└─ contacts/
  |  |└─ billing/
  |  |└─ subscriptions/
  |  └─ notifications/
```

```
|   ├── audit/
|   └── health/
|
|   ├── common/
|   ├── database/
|   ├── queues/
|   ├── config/
|   └── main.ts
```

Infrastructure repo

Alag rakho:

```
infra/
|   ├── docker/
|   |   ├── postfix/
|   |   ├── dovecot/
|   |   ├── rspamd/
|   |   ├── clamav/
|   |   ├── roundcube/
|   |   └── nginx/
|   |
|   ├── dns/
|   ├── scripts/
|   ├── backups/
|   ├── monitoring/
|   └── docker-compose.yml
```

Production mein later:

```
k8s/
|   ├── api/
|   ├── web/
|   ├── workers/
|   ├── redis/
|   ├── postgres/
|   ├── roundcube/
|   └── monitoring/
```

Lekin starting mein Kubernetes mat lagana.

developers ka exact division

Developer 1 — Backend + Mail Infrastructure

Primary:

Node.js/NestJS

PostgreSQL

Redis

Postfix

Dovecot

Rspamd

ClamAV

DKIM

DNS

Roundcube integration

DevOps

Daily responsibility:

API

DB

Mail server

Authentication

Mailbox provisioning

DNS

Security

Deployment

Developer 2 — Next.js + UI

Primary:

Next.js

React

TypeScript

Tailwind

Admin UI

Dashboard

Settings

Calendar UI

Contacts UI

Billing UI

Roundcube side:

skin

plugins

SSO/integration

branding

PHASE 0 — Foundation

Days 1–3

Developer 1:

PostgreSQL

Redis

Node

NestJS

Docker

Git

CI/CD

Environment

Developer 2:

Next.js

TypeScript

Tailwind

UI system
Layout
Auth pages

Output:

Frontend running
Backend running
DB running
Redis running
CI running

PHASE 1 — Authentication + Tenant

Days 4–8

Build:

Registration
Login
Logout
Refresh token
Password reset
Email verification
Tenant
User
Roles
Permissions

Roles:

SUPER_ADMIN
TENANT_ADMIN
USER

PHASE 2 — Domain + Mailbox Management

Days 9–14

Admin:

Add Domain
Verify Domain
Create Mailbox
Delete Mailbox
Suspend Mailbox
Change Password
Set Quota
Create Alias
Delete Alias

Example:

Domain
example.com

Mailbox
john@example.com

Quota
10 GB

Backend automatically:

PostgreSQL
↓
Provision mailbox
↓
Dovecot
↓
Mail storage

PHASE 3 — Roundcube

Days 15–18

Install:

Roundcube
PHP
Nginx
PostgreSQL

Configure:

IMAP → Dovecot

SMTP → Postfix

Roundcube's official requirements include PHP 8.1+, a supported DB, and IMAP/SMTP servers.

Then:

Next.js

↓

Open Mail

↓

Roundcube

Brand:

Your Logo

Your Name

Your Colors

Your Domain

PHASE 4 — Mail Security

Days 19–25

Setup:

SPF

DKIM

DMARC

PTR

TLS

Rspamd

ClamAV

Fail2ban

Rate limiting

Mail score:

Incoming

↓

Rspamd

↓

Spam score

↓

Reject / Accept / Spam

Rspamd itself is free/open-source, so there is no software license fee for normal use.

PHASE 5 — Business Features

Days 26–35

Next.js:

Dashboard

Storage usage

Mailbox settings

Aliases

Forwarding

Auto responder

Security

2FA

Sessions

Activity logs

Backend:

2FA

Audit logs

Notifications

Queue

Rate limits

PHASE 6 — Calendar + Contacts

Days 36–45

Calendar:

- Calendar
- Create event
- Edit
- Delete
- Invite
- Accept
- Reject
- Recurring event
- Reminders

Contacts:

- Contacts
- Groups
- Import
- Export
- Search
- Autocomplete

Iske liye Radicale/CalDAV/CardDAV use kar sakte ho.

PHASE 7 — Advanced Mail Features

Days 46–60

Implement:

- Schedule Send
- Undo Send
- Snooze
- Email templates
- Migration
- Sieve filters
- Auto forwarding
- Vacation responder

Migration:

- Gmail
- Office365
- IMAP

↓
Migration Job
↓
IMAP sync
↓
New mailbox

Queue:

BullMQ
↓
Migration Worker

PHASE 8 — Billing + Production

Days 61–70+

Implement:

Plans
Subscriptions
Payment gateway
Invoice
Usage limits
Storage limits
Mailbox limits
Billing webhooks

Then:

Monitoring
Backup
Restore
Security audit
Load testing
Mail deliverability
Production deployment

Actual monthly cost

Agar VPS se start karoge:

Development

1 VPS

2–4 CPU

8 GB RAM

100–160 GB SSD

~₹800–₹2,000/month

But mail ke liye main production mein single server recommend nahi karunga.

Small production

MAIL SERVER

4 CPU

8–16 GB RAM

200+ GB SSD

APP SERVER

4 CPU

8 GB RAM

100 GB SSD

DB/Redis

4 CPU

8 GB RAM

100 GB SSD

Roughly: ₹5,000 – ₹15,000/month

provider, disk, bandwidth, backup aur region par depend karega.

Security architecture

HTTPS

TLS

JWT/session security

2FA

Rate limiting

Fail2ban

Firewall

SSH keys

No root login

Secrets management

Encrypted backups

Audit logs

API:

Next.js

↓ HTTPS

Node API

↓

Auth

↓

RBAC

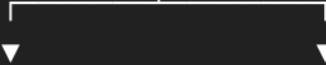
FRONTEND

Next.js + React + TypeScript + Tailwind



BACKEND

Node.js + NestJS + REST + WebSocket



PostgreSQL

Redis



BullMQ



MAIL INFRASTRUCTURE

Postfix → SMTP

Dovecot → IMAP

Rspamd → Spam

ClamAV → Virus

DKIM → Signing

Roundcube → Webmail